Software Requirements Specification

A Mobile Application for Facility Management

Team
[SEBI]

Members
[Abdulrahman Mohamed] **|** [13004274] (Team Lead)
[Selim Omar] **|** [13001408]
[Fares Fares] **|** [13006971]
[Abdullah Atef] **|** [13005088]
[Fares Shady] **|** [13005225]
[Amir Ali] **|** [13007425]

Tutorial Details
[Tutorial 4] **|** [Hassan]

1. Introduction

This document presents the Software Requirements Specification (SRS) for a mobile-based Facility Management System designed specifically for the **German International University (GIU) community**.
The system aims to support students, staff, and facility personnel at GIU by providing a digital platform to report and manage maintenance issues across the campus.
The intended audience includes developers, project stakeholders, and university staff involved in improving campus services.
The application enhances communication between GIU community members, facility managers, and workers by streamlining the issue reporting and resolution process.

1.1 Product Vision & Scope

The goal of this system is to provide a centralized platform for the **GIU community** where users can easily report facility-related issues within the university campus and track their progress.

The system is designed to improve campus maintenance efficiency by enabling faster communication between:

- GIU students and staff (Community Members)
- Facility Managers
- Maintenance Workers

The application will:

- Allow GIU users to report issues occurring on campus
- Improve response time for maintenance tasks
- Increase transparency by allowing users to track issue status
- Support better coordination between facility teams

1.2 Definitions and Acronyms

SRS: Software Requirements Specification
Ticket: A digital record representing a maintenance issue
RBAC: Role-Based Access Control

2. Functional Requirements

1. Community Member Functional Requirements

FR-CM-01: Account Creation
The system shall allow a Community Member to create an account by providing required information (e.g., name, email, and password).

FR-CM-02: User Authentication
The system shall allow a Community Member to log in using a valid email and password.

FR-CM-03: Issue Submission
The system shall allow a Community Member to submit an issue including the following details:

- Location
- Photo
- Category
- Description

FR-CM-04: Issue Description
The system shall allow a Community Member to add a textual description when submitting an issue.

FR-CM-05: View Submitted Issues
The system shall allow a Community Member to view a list of issues they have submitted.

FR-CM-06: Logout
The system shall allow a Community Member to log out of the application.

FR-CM-07: View Issue Status
The system shall display the status of an issue to the Community Member.
Possible statuses include:

- Pending

- In Progress
- Resolved

2. Facility Manager Functional Requirements

FR-FM-01: Account Creation
The system shall allow a Facility Manager to create an account.

FR-FM-02: User Authentication
The system shall allow a Facility Manager to log in using a valid email and password.

FR-FM-03: Logout
The system shall allow a Facility Manager to log out of the application.

FR-FM-04: View All Issues
The system shall allow a Facility Manager to view all submitted issues in the system.

FR-FM-05: Update Issue Status
The system shall allow a Facility Manager to update the status of an issue.

FR-FM-06: Assign Issue to Worker
The system shall allow a Facility Manager to assign an issue to a specific worker.

FR-FM-07: Close Issue
The system shall allow a Facility Manager to close an issue once it has been resolved.

3. Worker Functional Requirements

FR-W-01: Account Creation
The system shall allow a Worker to create an account.

FR-W-02: User Authentication
The system shall allow a Worker to log in using a valid email and password.

FR-W-03: Logout
The system shall allow a Worker to log out of the application.

FR-W-04: View Assigned Issues
The system shall allow a Worker to view issues assigned to them.

FR-W-05: Comment on Issue
The system shall allow a Worker to add comments to an issue.

FR-W-06: Update Issue Status to In Progress
The system shall allow a Worker to mark an assigned issue as **In Progress**.

FR-W-07: Upload Completion Photo
The system shall allow a Worker to upload a photo after completing the work on an issue.

4. System Admin Functional Requirements (Optional)

FR-SA-01: Admin Authentication
The system shall allow a System Admin to log in.

FR-SA-02: Logout
The system shall allow a System Admin to log out.

FR-SA-03: Account Management
The system shall allow a System Admin to activate or deactivate user accounts.
FR-SA-04: View Registered Users
The system shall allow a System Admin to view all registered users in the system.

2.1 User stories:

A GIU community member user story:

- A student notices a trash can in the bathroom is overflowing, she checks the list of tickets and then submits a ticket by taking a photo, choosing the location and entering additional information. She receives a notification that it is submitted and another notification when the ticket's status is marked as resolved.

- A facility manager logs into the system and views all submitted tickets across the GIU campus. She notices a newly reported issue about an overflowing trash can. She reviews the issue details, including the photo and location, and assigns it to an available worker. As the worker updates the ticket status to "In Progress" and later completes the task, the facility manager verifies the work using the uploaded completion photo and marks the ticket as resolved.

- A worker logs into the system and checks the list of issues assigned to him. He sees a ticket about an overflowing trash can in one of the campus bathrooms. He updates the status to "In Progress" to indicate that he has started working on it. After cleaning and fixing the issue, he uploads a photo as proof and adds a comment confirming completion. The ticket is then ready for the facility manager to review.

- A system admin logs into the system to manage users within the GIU facility management application. She reviews the list of registered users and notices that one account is inactive or violating system policies. She deactivates the account to maintain system integrity. Additionally, she ensures that all users have appropriate roles assigned and that the system is functioning correctly.

1. Community Member

US-CM-01:
As a Community Member, I should be able to create an account so that I can log in to the application.

US-CM-02:
As a Community Member, I should be able to log in using my email and password so that I can access the application

US-CM-03:
As a Community Member, I should be able to submit an issue with its details (location, photo, and category) so that the problem can be reported accurately.

US-CM-04:
As a Community Member, I should be able to add a description to the issue so that the worker understands the problem clearly.

US-CM-05:
As a Community Member, I should be able to view my submitted issues so that I can track their status.

US-CM-06:
As a Community Member, I should be able to log out so that my data remains secure on shared devices.

US-CM-07
As a Community Member, I should be able to view the status of my issue (Pending, In Progress, Resolved) so that I know the progress of my request.

2. Facility Manager

US-FM-01:
As a Facility Manager, I should be able to create an account so that I can log in to the application.

US-FM-02:
As a Facility Manager, I should be able to log in using my email and password so that I can access the application.

US-FM-03:
As a Facility Manager, I should be able to log out so that my data remains secure on shared devices.

US-FM-04:
As a Facility Manager, I should be able to view all submitted issues so that I can assign them to workers.

US-FM-05:
As a Facility Manager, I should be able to update an issue's status so that the system reflects the current state of the issue.

US-FM-06
As a Facility Manager, I should be able to assign an issue to a worker so that the worker can start fixing it.

US-FM-07
As a Facility Manager, I should be able to close an issue after it is resolved so that the system reflects that the issue is completed.

3. Worker

US-W-01:
As a Worker, I should be able to create an account so that I can log in to the application.

US-W-02:
As a Worker, I should be able to log in using my email and password so that I can access the application.

US-W-03:
As a Worker, I should be able to log out so that my data remains secure on shared devices.

US-W-04:
As a Worker, I should be able to view my assigned issues so that I can start working on them.

US-W-05:
As a Worker, I should be able to comment on an issue to indicate that the work is completed so that the facility manager can update its status.

US-W-06
As a Worker, I should be able to mark an issue as “In Progress” so that the facility manager and community member know that it is being worked on.

US-W-07
As a Worker, I should be able to upload a photo after fixing the issue so that the facility manager can verify the work.

4. System Admin:

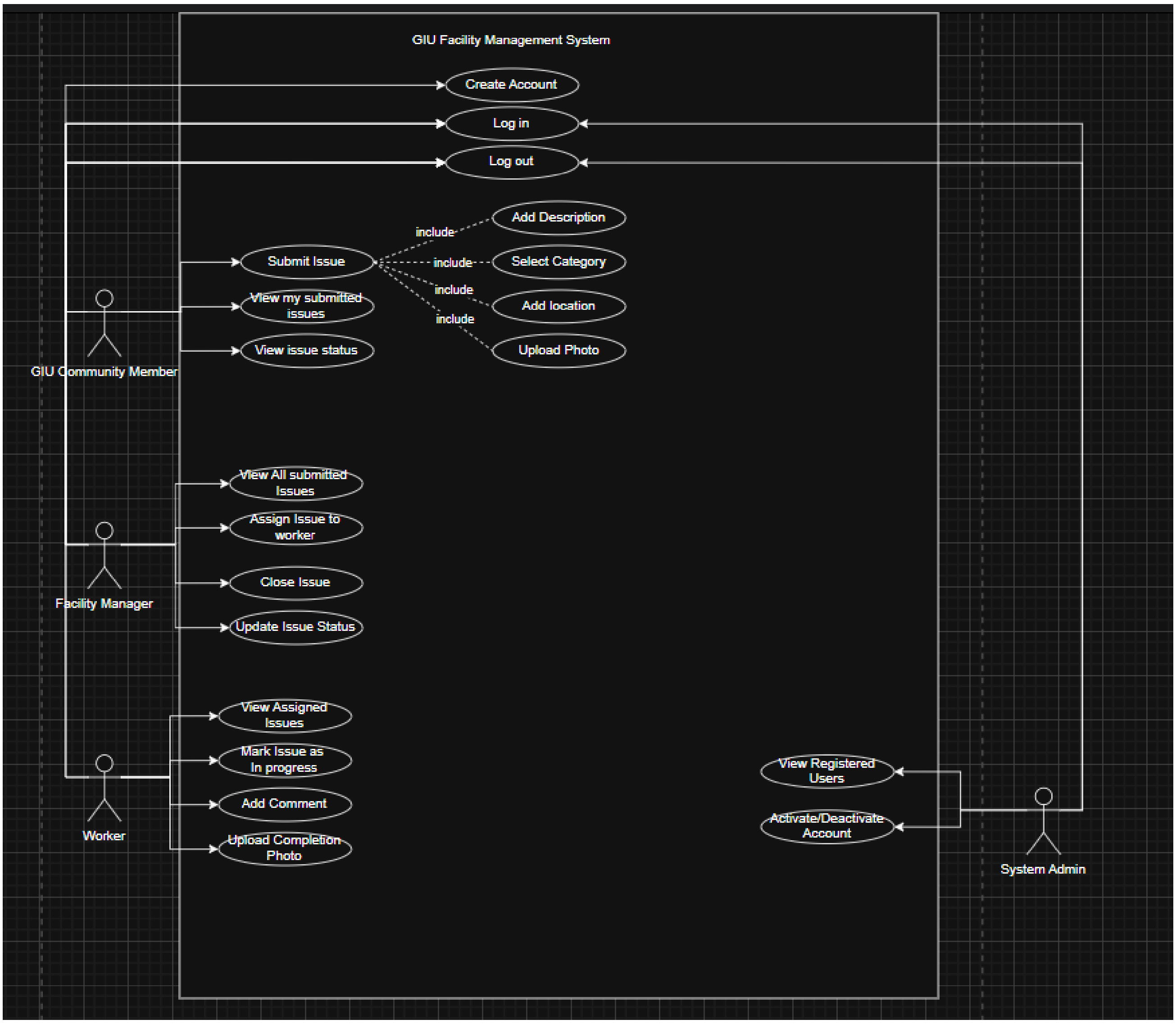
US-SA-01:
As a System Admin, I should be able to log in so that I can access the application.

US-SA-02:
As a System Admin, I should be able to log out so that my data remains secure on shared devices.

US-SA-03:
As a System Admin, I should be able to activate or deactivate any account so that I can ensure all users follow the system rules.

US-SA-04
As a System Admin, I should be able to view all registered users so that I can manage the system users

2.2 UML Use-case Diagram



3. Non-functional Requirements:

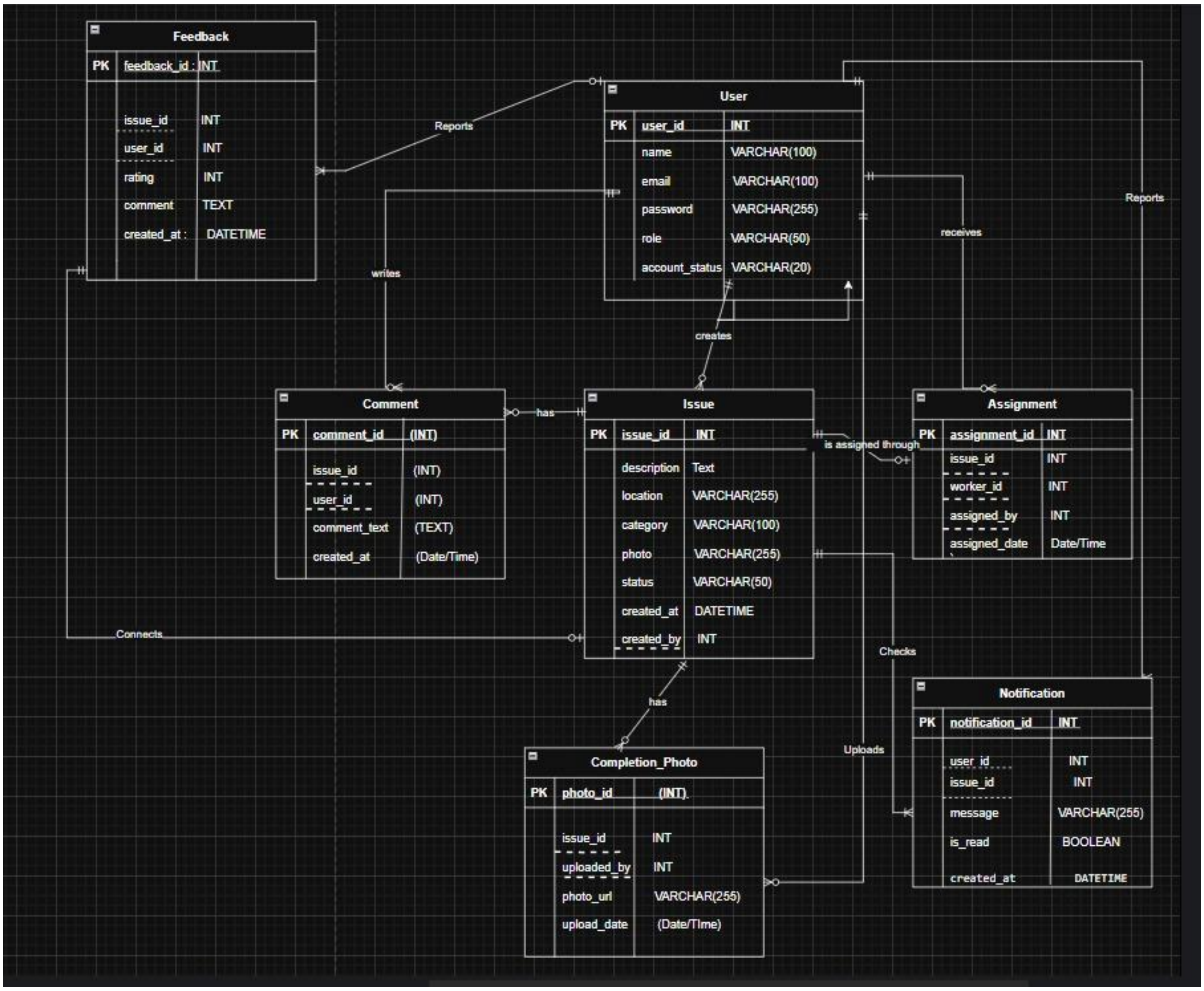
Details the performance, usability, security, and other constraints of the system.

- Response Time: Page load times under three seconds
- Concurrent Users: Support for at least 500 simultaneous users
- API Security: Rate limiting and input validation
- Backup & Recovery: Automated daily backups with disaster recovery plan
- Usability: Intuitive UI such that tickets for issues can be submitted in under one minute.
- Reliability: Must handle image uploads robustly, even on unstable network connections.

4. Constraints

- The system is designed specifically for use within the **GIU campus environment**.
- Requires internet connectivity within campus or mobile networks.
- Limited development time due to semester constraints.
- The system depends on user devices for capturing and uploading images.

5. Entity Relation Diagram



6. Technology Stack

- Response Time: Page load times under three seconds
- Concurrent Users: Support for at least 500 simultaneous users
- API Security: Rate limiting and input validation
- Backup & Recovery: Automated daily backups with disaster recovery plan

7. Future Scope of the Project

For the future scope of this project, we are hoping to add 2 functionalities:

- The first functionality is Feedback. The GIU Community members will be able to leave a rating out of 5 and a small text to rate/give their feedback on their submitted issue.

- The second functionality will be that the GIU Community member will receive a notification telling them that their issue has been resolved.